

Linear approximations of large deviations: cubic diffusion test.

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Abstract

Large deviation techniques have been widely used in physics to study the fluctuations of nonequilibrium systems, which are often modelled using Markov processes, including Markov diffusion describe by stochastic differential equations (SDEs). Large deviation theory allows for the calculation of the probability distribution of time integrate of the state, which can represent observables such the work done on a Brownian particle or the heat exchanged with the environment. In many cases, obtaining exact analytical solutions of large deviation functions requires solving complex spectral or optimization problems [1, 2]. This same complexity arises when trying to understand how fluctuations arise in effective or driven processes. In these systems, fluctuations are viewed as optimal modifications to the stationary density or as an effective diffusion process that adjusts the original force or drift of the diffusion [2, 3, 4].

I will present in this talk, a method for approximating the large deviation rate function of time-integrated observables of diffusion processes, used in statistical physics to characterize the fluctuations of nonequilibrium systems. The method is based on linearizing the effective process associated with the large deviations of the process and observable considered, and is tested for a simple one-dimensional nonlinear diffusion model involving a cubic drift. The results show that the linear approximation compares well with the exact rate function, especially in the large fluctuation regime, and that its accuracy is related to the way the linearized process localizes in space. Possible extensions and applications to more complex diffusion models are proposed for future work.

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